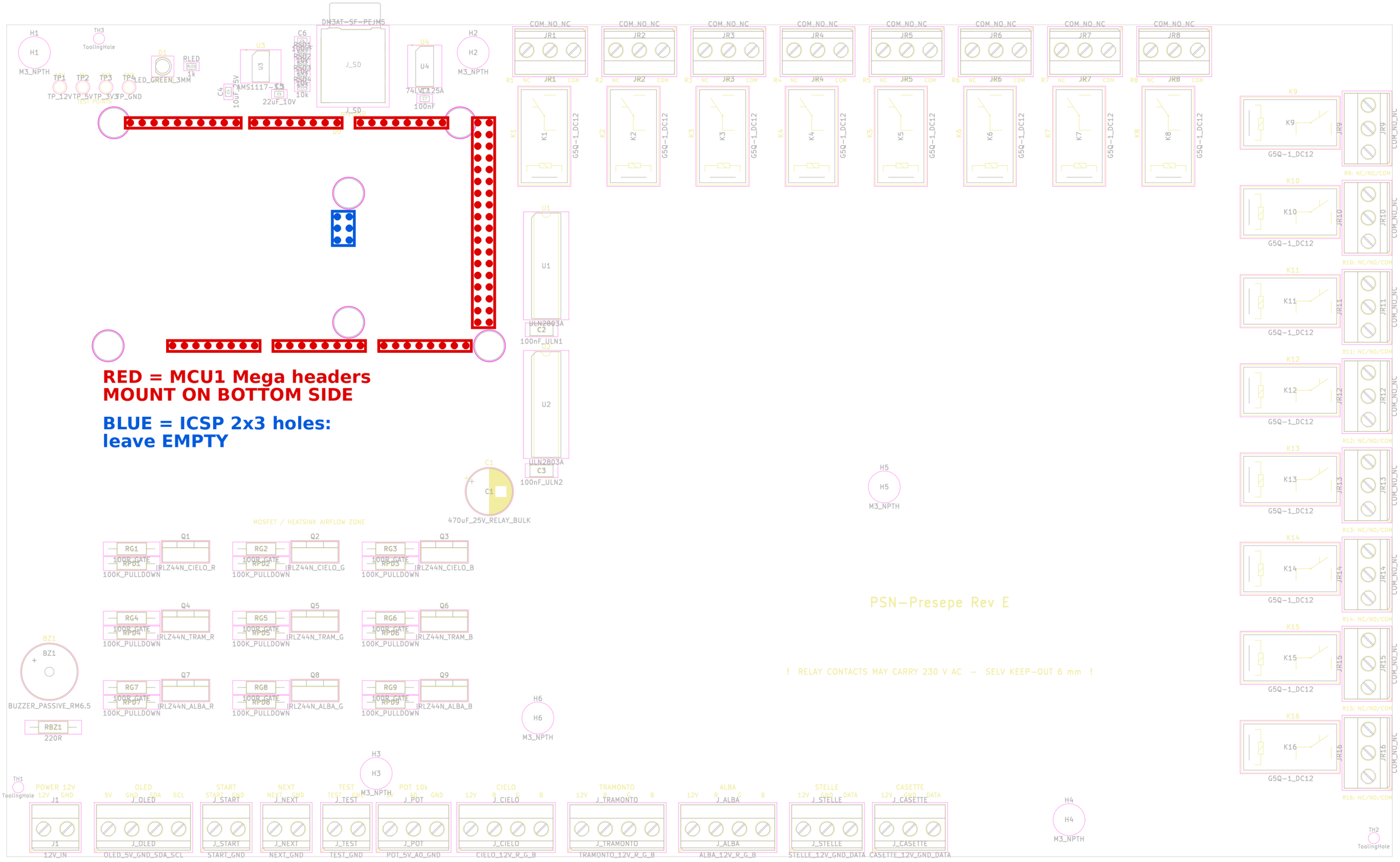
PSN-Presepe Rev E - Assembly drawing (TOP view)



PSN-Presepe Rev E - Assembly instructions

1. Turnkey assembly, all parts per BOM (PSN-Presepe-Mega-RevE-BOM-PCBWay.csv). Centroid: PSN-Presepe-Mega-RevE-Centroid-PCBWay.csv.
2. TOP side: all SMD parts (13) and all through-hole parts except MCU1.
3. MCU1 = Arduino Mega 2560 R3 shield male pin headers (2.54 mm): 1x10, 5x 1x8, 1x 2x18 = 86 pins (red on page 1).
   MOUNT ON THE BOTTOM SIDE: plastic body under the board, long pins pointing DOWN (they plug into an Arduino Mega).
   Solder on the TOP side. Pins must be straight and perpendicular: use an Arduino Mega board as alignment jig.
4. ICSP 2x3 holes (blue on page 1): leave EMPTY, no header.
5. Polarity: C1 + = square pad (+ on silkscreen). BZ1 + mark = square pad, pins 6.5 mm apart: insert straight.
   D1 (3 mm LED): cathode = short lead / flat side = square pad. U1/U2 (ULN2803A DIP-18) pin-1 notch as silkscreen.
   K1-K16 (Omron G5Q-1 DC12) pin 1 = square pad (coil +12V). Q1-Q9 IRLZ44N: metal tab towards silkscreen tab line.
6. J_SD (Hirose DM3AT-SF-PEJM5 microSD push-push): card opening faces the top board edge.
7. Screw terminals (Phoenix MKDS 1,5 5.08 mm): wire entry faces the nearest board edge.
8. Relay contacts may carry 230 VAC: do NOT add wires, jumpers, vias or copper. Clean flux residues, especially in the relay area.
9. Mounting holes H1-H6 and tooling holes TH1-TH3 are NPTH: leave empty. Test points TP1-TP4: bare pads, no part.
10. No IC programming required.
11. Please send photos of the first assembled board (top and bottom) before shipping.